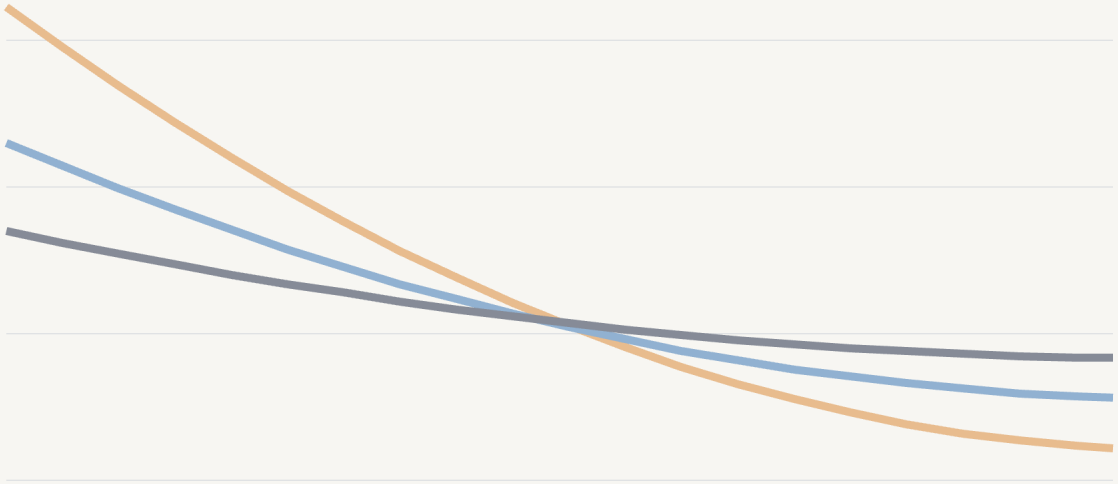
FINANCIAL ENGINEERING

The Volatility Surface

How one object describes the collective movement of option prices



Why quote option prices in volatility

INPUT

Option price C(K, T)

Not comparable across strikes and expiries

→

TRANSFORM

Invert Black–Scholes

Strips out the mechanical effect of S, K, T and r

→

OUTPUT

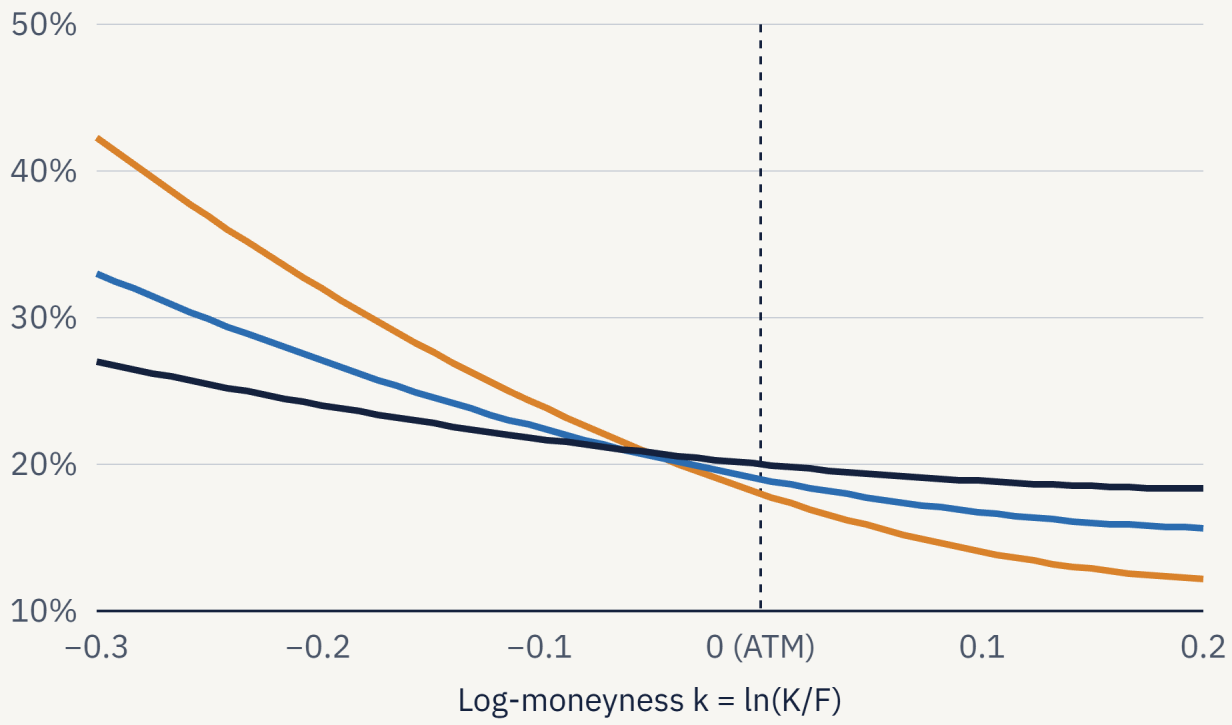
Implied vol σ(K, T)

One comparable unit for every option

If Black–Scholes were true, the surface would be flat: one vol for every strike and expiry.

Its actual shape is a map of where, and how much, the market disagrees with lognormal dynamics.

Strike dimension: the smile



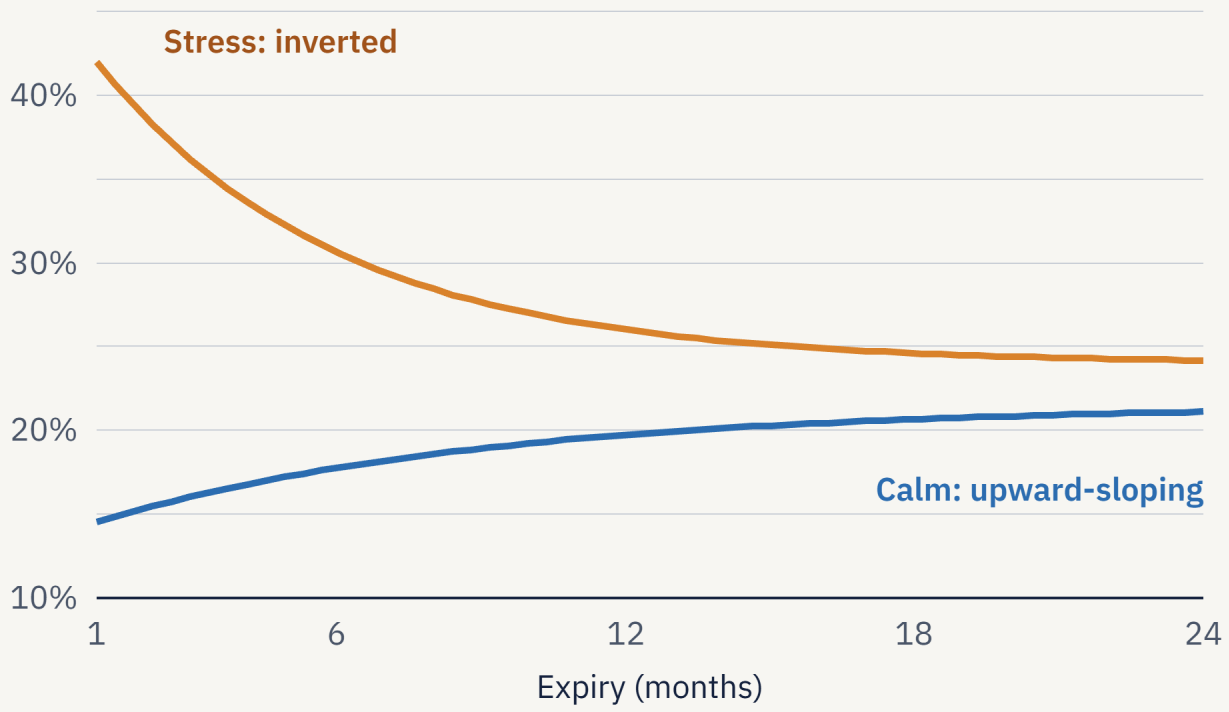
——— 1 month (steepest)
——— 3 months
——— 1 year (flattest)

Equity indices: downside puts trade rich, reflecting crash risk and negative spot–vol correlation.

Skew flattens with maturity, roughly $\propto 1/\sqrt{\tau}$, as jumps average out.

$\partial^2 C/\partial K^2$ is the discounted risk-neutral density: the smile is the implied distribution.

Maturity dimension: the term structure



ATM vol by expiry ≈ expected realized vol over each horizon plus a variance risk premium.

Calm markets: upward-sloping. **Stress:** inverted, as front-month vol spikes.

The front end is far more reactive, which is why desks track weighted vega, not raw vega.

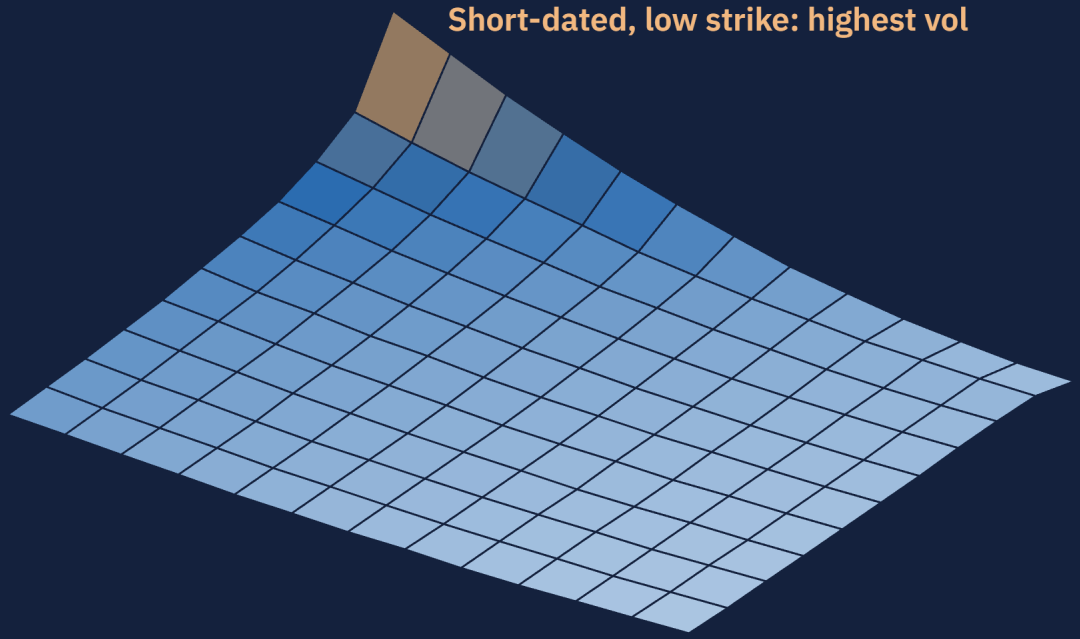
Put together: one surface σ(k, τ)

One implied vol for every strike and expiry, fitted into a smooth function.

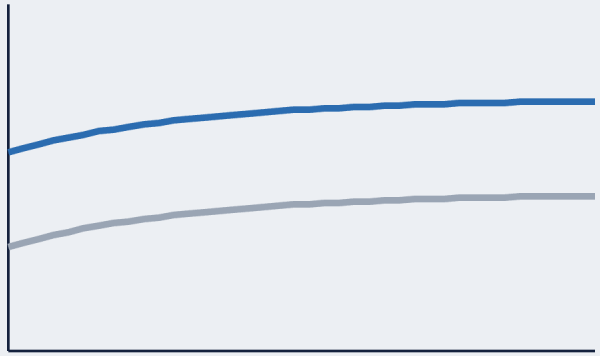
Back edge: short-dated smiles, steep, with the highest vols at low strikes.

Front edge: long-dated smiles, shallow and close to flat.

Neighbors move together: option prices change collectively, not one by one.

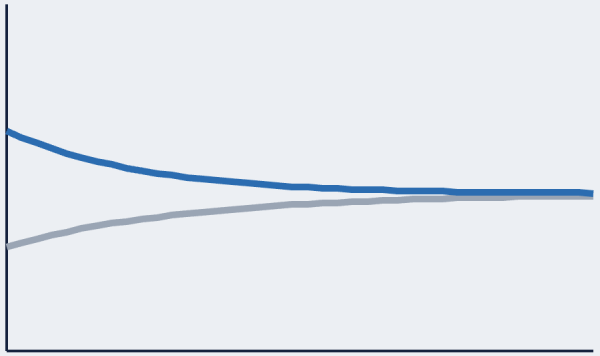


Three factors drive the whole surface



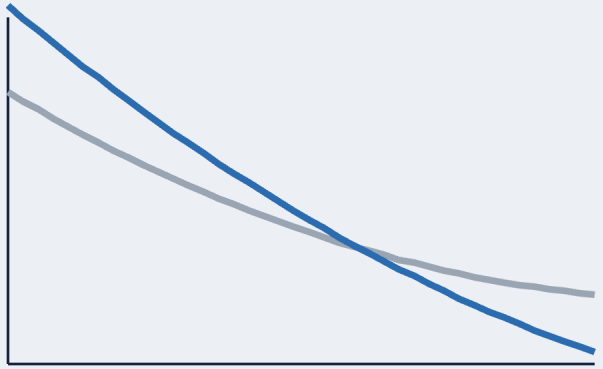
1. Level

Everything shifts together.
Dominant, hence aggregate vega.



2. Term tilt

Short-dated vol moves far more than long-dated.



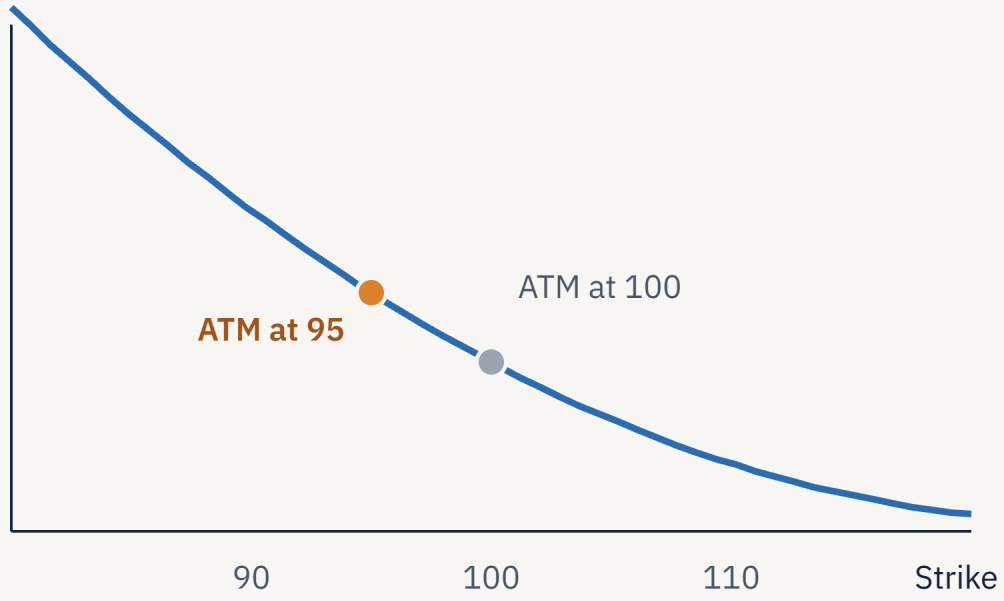
3. Skew

The smile steepens or flattens; wings move vs. ATM.

Grey = before, blue = after. PCA on daily surface changes: together these typically explain **85–95% of the variance**.

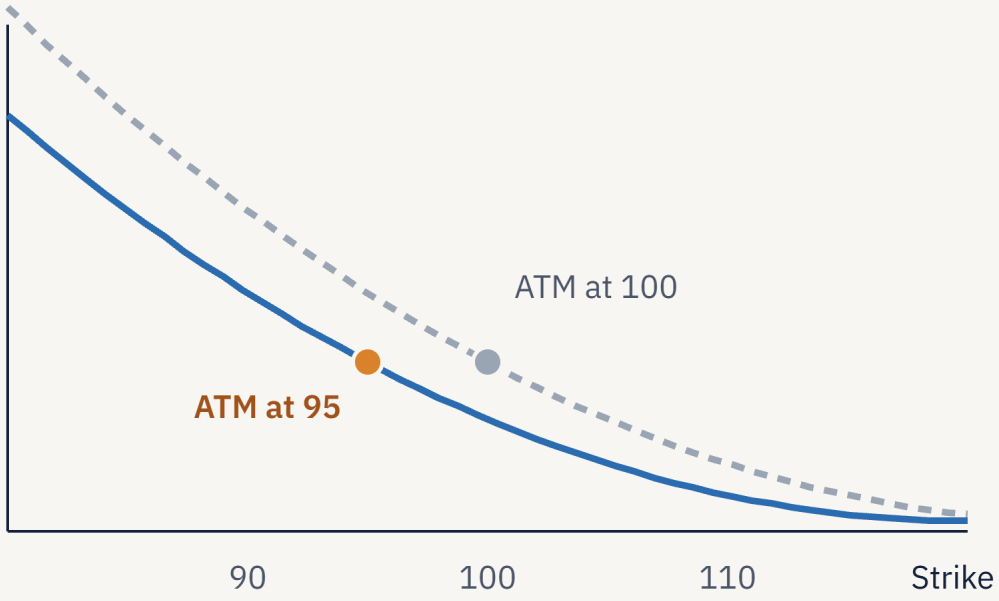
When spot falls from 100 to 95

Sticky strike



Each strike keeps its vol; ATM vol rises as spot slides down the skew.

Sticky moneyness



The smile shifts with spot; ATM vol holds, fixed-strike vols move.

Smile-adjusted $\Delta = \partial V/\partial S + \text{vega} \cdot \partial\sigma/\partial S$: the regime you assume changes your hedge.

Keeping the surface arbitrage-free

Condition	Requirement	If violated
Vertical spread	Calls decrease in strike: $\partial C/\partial K \le 0$	A call spread with negative cost
Butterfly	Calls convex in strike: $\partial^2 C/\partial K^2 \ge 0$	A negative implied density
Calendar	Total variance $w = \sigma^2\tau$ non-decreasing in τ at fixed moneyness	A calendar spread that pays you to own it

Common parameterizations

SVI / SSVI · equity indices

SABR · rates and FX

Vanna–volga · FX wings

The fitted parameters (level, skew, curvature, wings) become a compact state vector. **Their daily changes are the collective movement.**

Putting the surface to work

Marking exotics

Priced off local-vol or SLV models calibrated to the whole surface.

Risk

Vega bucketed by expiry and strike; scenarios built from the three factors.

Relative value

Rich/cheap signals from dislocations against the fit or its own history.

Density extraction

Each expiry implies a risk-neutral density; wing fits drive its stability.

Level, term tilt and skew, plus how the surface co-moves with spot, explain most of what an options book does.

The moment surface

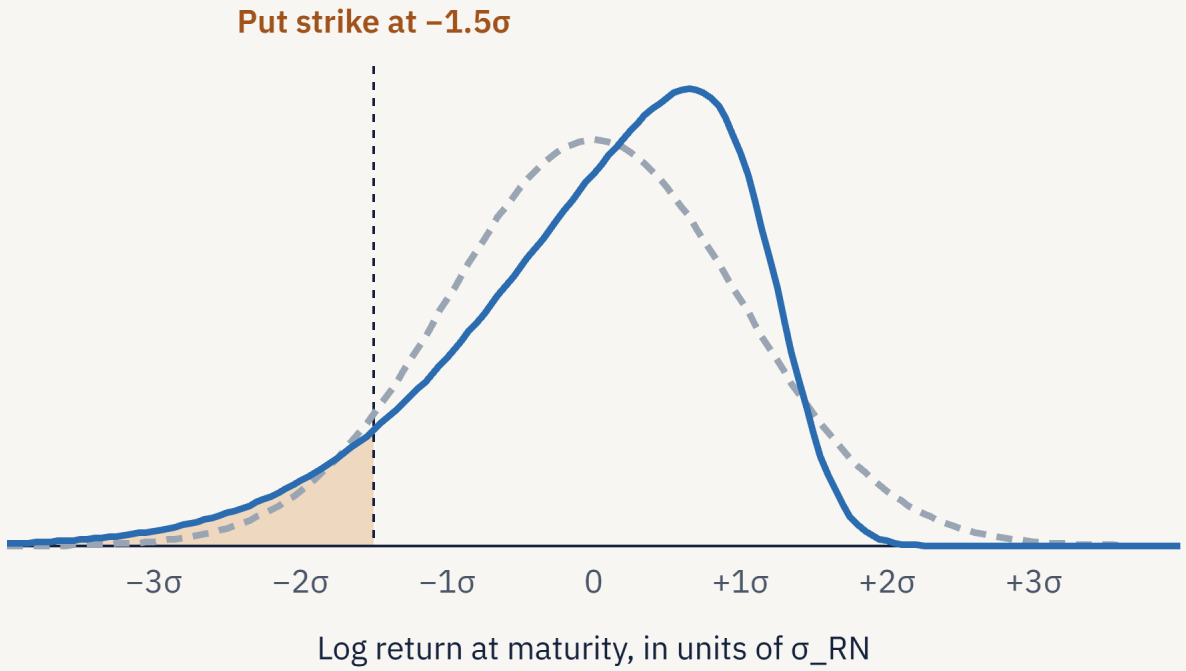
Replace a strike-indexed vol vector with properties of the risk-neutral distribution.

MODEL-FREE $$E_Q[R^2] = \int 2(1 - \ln(K/S)) / K^2 \cdot Q(K)\, dK$$

Moment per expiry	From the OTM strip	What it measures
$\sigma_{RN}(T) = \sqrt{(Var_Q[R] / T)}$	Weights on Q(K) for R^2	Dispersion of the RN distribution
$Skew_{RN}(T)$	Same strip, weights for R^3	Asymmetry: crash vs. rally pricing
$Kurt_{RN}(T)$	Same strip, weights for R^4	Tail weight

R = ln(S_T/S); Q(K) = OTM option price at strike K (Bakshi, Kapadia & Madan, 2003). Not the VIX: the log contract equals integrated variance only without jumps.

Why σ alone isn't enough



– – – Symmetric
——— Skewed (skew ≈ −1.7)

Same mean, same σ_RN.

The −1.5σ put is worth **about 1.9×** as much under the skewed density.

σ_RN can't reprice the smile on its own. **Add skewness and kurtosis.**

Fewer numbers, each with a meaning

IMPLIED VOL SURFACE

$15 \times 8 = 120$

Strikes × expiries, no shared meaning

→

MOMENT SURFACE

$3 \times 8 = 24$

σ, skew, kurtosis per expiry

Surface PCA factor	Moment counterpart
Level and term tilt	σ_RN term structure
Skew	Risk-neutral skewness by expiry
Curvature and wings	Risk-neutral kurtosis by expiry

To reprice strikes, fit a parametric density per expiry (lognormal mixture, Edgeworth, generalized beta): few parameters, distributional meaning, a full price surface.

What to watch in practice

⚠ Tails dominate

Truncating at the last listed strike biases σ_RN down, worse for skew and kurtosis (Jiang & Tian, 2005).

⚙ Noisy wings

Fit an arbitrage-free smile first, then integrate. Calibration survives only as interpolation.

🔑 Still risk-neutral

σ_RN includes the variance risk premium: priced dispersion, not a forecast of realized.

The same information as the surface, organized as moments of the distribution: interpretable, compact, and aligned with how surfaces actually move.